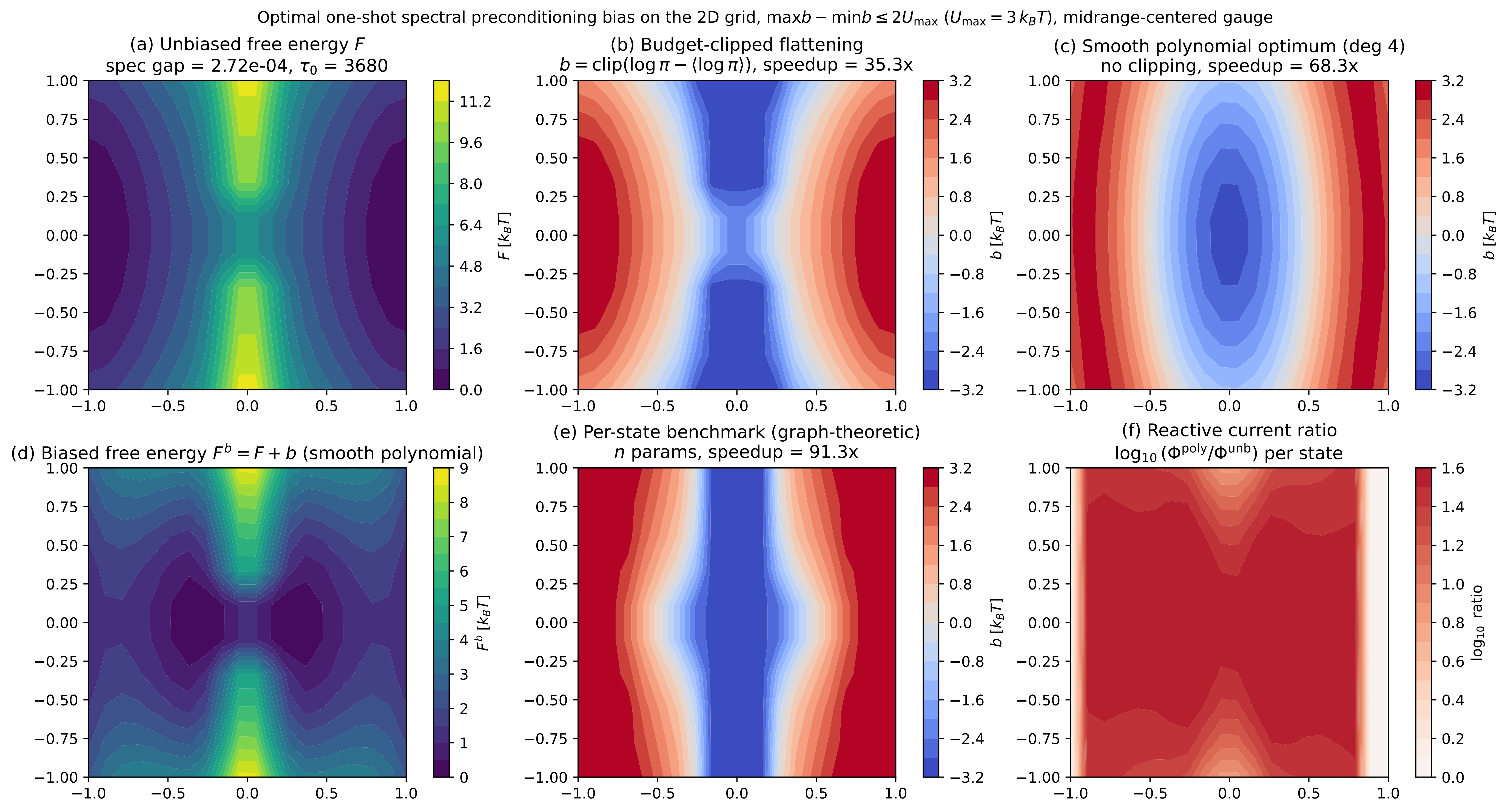

Optimal one-shot spectral preconditioning bias on the 2D grid, $\max b - \min b \le 2U_{\max}$ ($U_{\max} = 3\,k_BT$), midrange-centered gauge
(a) Unbiased free energy F
spec gap = 2.72e-04, τ_0 = 3680
F [k_BT]
(b) Budget-clipped flattening
$b = \mathrm{clip}(\log\pi - \langle\log\pi\rangle)$, speedup = 35.3x
b [k_BT]
(c) Smooth polynomial optimum (deg 4)
no clipping, speedup = 68.3x
b [k_BT]
(d) Biased free energy $F^b = F + b$ (smooth polynomial)
F^b [k_BT]
(e) Per-state benchmark (graph-theoretic)
n params, speedup = 91.3x
b [k_BT]
(f) Reactive current ratio
$\log_{10}(\Phi^{\mathrm{poly}}/\Phi^{\mathrm{unb}})$ per state
$\log_{10}$ ratio